

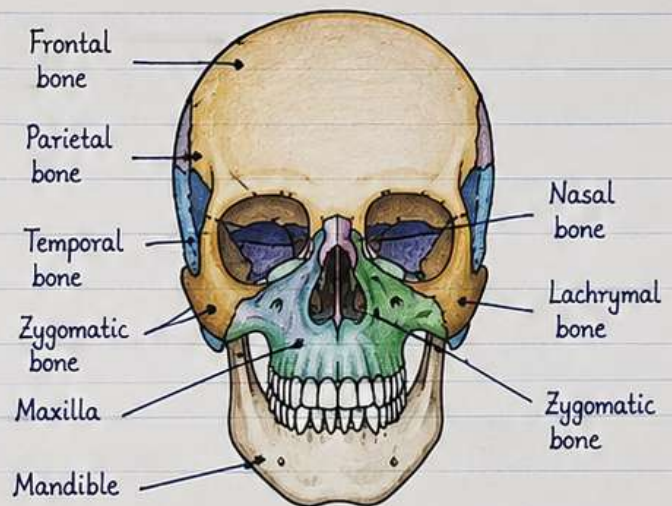
1. INTRODUCTION

After studying this section, you should be able to :

- identify the bones of the skull (face and cranium).
- list the functions of the sinuses and fontanelles of the skull.
- outline the characteristics of a typical vertebra.
- describe the structure of the vertebral column.
- explain the movements and functions of the vertebral column.
- identify the bones forming the thoracic cage.

The bones of the skeleton are divided into two groups : axial skeleton and appendicular skeleton (Fig. 16.13).

The axial skeleton, so-called because it is the axis or central core of the skeleton, consists of the skull, vertebral column, ribs and sternum. The appendicular skeleton consists of the shoulder and pelvic girdles and the limb bones.

2. SKULL

The skull (Figs. 16.14 and 16.15) rests on the upper end of the vertebral column and its bony structure is divided into two parts :

- (i) Cranium (ii) Face

(i) CRANIUM

The cranium is formed by several flat and irregular bones that protect the brain. The brain rests on the base of the skull (Fig. 16.15) and is enclosed in the cranial cavity.

The prominent lining the inner surface of the skull bones forms the outer layer of dura mater (p.167). In the mature skull the joints (sutures) between the bones are immovable. The bones have numerous perforations (e.g. foramina, fissures) through which nerves, blood and lymph vessels pass.

Fig. 16.16 : The bones of the face.
(Anterior view)

Bones of the cranium are :

- 1 frontal bone
- 2 parietal bones
- 2 temporal bones
- 1 occipital bone
- 1 sphenoid bone
- 1 ethmoid bone

3. FACIAL BONES

The skeleton of the face is formed by 13 bones in addition to the 6 ear ossicles and the hyoid bone. These bones fit together like an intricate, three-dimensional jigsaw puzzle.

The facial bones form the framework of the face, the walls of the orbits (eye sockets), the nasal cavity, the mouth and the upper jaw.

They are joined together by immovable joints (sutures), held firmly by ligaments and form a strong but light structure.

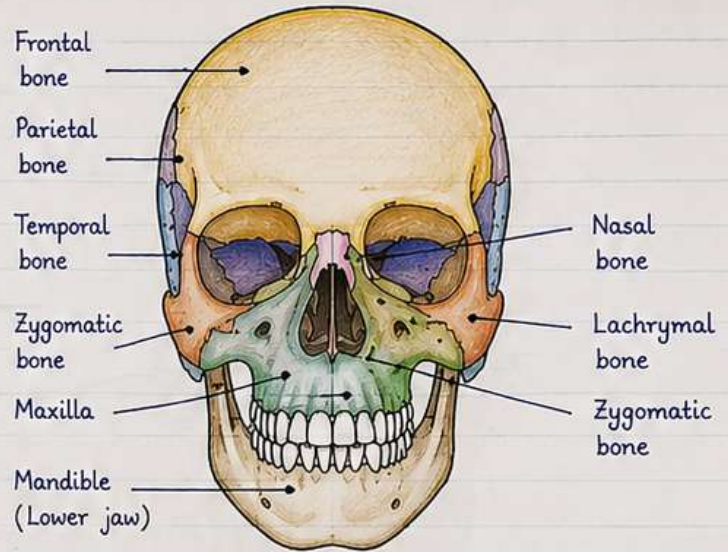


Fig. 16.16 : The bones of the face.
(Anterior view)

The facial bones are :

- | | |
|--|------------------------------------|
| (1) Pair of nasal bones | (5) Pair of palatine bones |
| (2) Pair of lacrymal bones | (6) Pair of inferior nasal conchae |
| (3) Pair of zygomatic bones | (7) Vomer |
| (4) Pair of maxillae (upper jaw bones) | (8) Mandible (lower jaw bone) |

Description of Facial Bones

1. Nasal bones (Pair)

- These two small flat bones form most of the upper and lateral part of the bridge of the nose.
- They give shape to the nose.

2. Lacrymal bones (Pair)

- These two small bones are posterior and lateral to the nasal bones.
- They form part of the medial walls of the orbits.
- Each is pierced by a foramen through which the nasolacrimal duct passes.

3. Zygomatic bones (Pair)

- These bones form the prominences of the cheeks (cheek bones).
- They also form part of the floor and lateral walls of the orbits.

7. Vomer (Single)

- The vomer is a thin, flat bone.
- It extends upward from the middle of the hard palate to form most of the inferior part of the nasal septum.
- It articulates with the perpendicular plate of the ethmoid.

4. Maxillae (Pair)

- The maxillae form the upper jaw, the anterior part of the roof of the mouth, the lateral walls of the nasal cavity and part of the floor of the orbits.
- The alveolar ridge (on maxilla) bears the upper teeth.
- Each side has a large maxillary sinus that opens into the nasal cavity.

5. Palatine bones (Pair)

- These are two small L-shaped bones.
- The horizontal part forms the posterior part of the hard palate.
- The perpendicular part forms part of the lateral wall of the nasal cavity and the floor of the orbit.

6. Inferior nasal conchae (Pair)

- Each is a scroll-shaped bone.
- They form part of the lateral wall of the nasal cavity.
- They increase the surface area and help in warming and humidifying the inspired air.

8. Mandible (Single)

- The mandible is the lower jaw bone.
- It is the only movable bone of the skull.
- It forms the lower boundary of the face and bears the lower teeth.

4. VERTEBRAL COLUMN

The vertebral column (Fig. 16.17) extends from the base of the skull to the tip of the tailbone. It is made up of 33 vertebrae arranged in five regions.

The vertebrae are separated by intervertebral discs which act as shock absorbers.

Regions of Vertebral Column :

- ① 7 cervical vertebrae - in the neck region.
- ② 12 thoracic vertebrae - in the chest region.
- ③ 5 lumbar vertebrae - in the abdomen region.
- ④ 5 sacral vertebrae (fused) - form the sacrum.
- ⑤ 4 coccygeal vertebrae (fused) - form the coccyx or tailbone.

Characteristics of a Typical Vertebra :

- Vertebrae are irregular bones.
- Each vertebra has a body in front and a ring-like arch behind.
- The arch has a central hole called vertebral foramen.
- The spinal cord passes through the vertebral foramen.
- Processes (projections) help in attachment of muscles.

Functions of Vertebral Column :

- Protects the spinal cord.
- Supports the head and trunk.
- Helps in maintaining posture and balance.
- Allows movements of the body such as bending, twisting, etc.
- Acts as a flexible structure due to intervertebral discs.

Vertebral column
(lateral view)

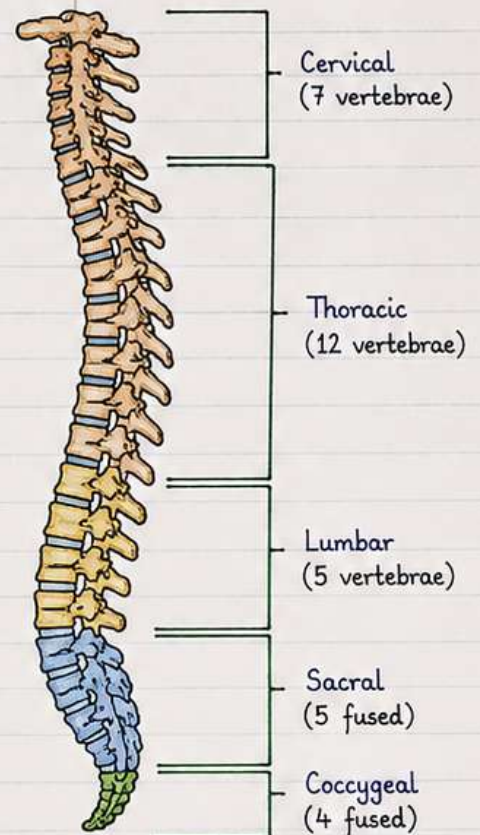
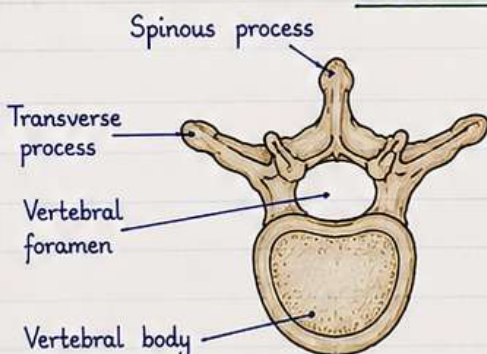


Fig. 16.17 : The vertebral column.

Intervertebral Discs :

These are pads of fibrocartilage present between two adjacent vertebrae. They act as shock absorbers and allow slight movement between vertebrae.

Structure of a Typical Vertebra (in brief)

- The vertebral body is the front, thick part which bears weight.
- The vertebral arch is the ring-like part behind the body.
- The vertebral foramen is the central hole through which the spinal cord passes.
- There are projections (processes) for attachment of muscles and ligaments.

5. LOWER PORTION OF THE HUMAN BODY

The lower limb (Fig. 16.18) consists of the hip bone, thigh bone (femur), kneecap (patella), lower leg bones (tibia and fibula) and foot bones (tarsals, metatarsals and phalanges).

(i) HIP BONE (Part of Pelvic Girdle)

- The hip bone is a large, curved bone.
- It forms the pelvis and joins with the thigh bone to form the hip joint.
- It protects the internal organs of the pelvic cavity.

(ii) FEMUR (Thigh Bone)

- Femur is the longest and strongest bone in the human body.
- It lies between the hip bone and the kneecap.
- Its upper end articulates with the hip bone to form the hip joint.

(iii) PATELLA (Kneecap)

- Patella is a small, flat, triangular bone.
- It lies in front of the knee joint.
- It protects the knee joint and increases the efficiency of the leg muscles.

(iv) TIBIA AND FIBULA (Lower Leg Bones)

- Tibia (Shin bone)
 - Tibia is the larger and stronger of the two bones.
 - It lies on the medial side of the leg.
 - It bears most of the body weight.
- Fibula (Slim bone)
 - Tibula is the thinner and weaker bone.
 - It lies on the lateral side of the tibia.
 - It does not bear much weight but provides attachment to muscles.

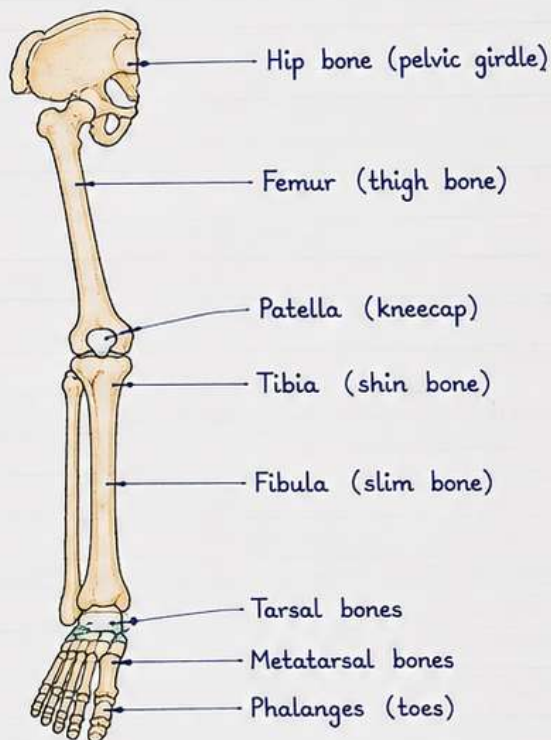


Fig. 16.18 : Bones of the lower limb (Right side, anterior view)

(v) BONES OF THE FOOT

The foot has 26 bones. They are grouped into three types :

(1) Tarsal bones (7)

- These are arranged in two rows.
- They form the ankle and give shape to the foot.
- The largest tarsal bone is the heel bone (calcaneus).

(2) Metatarsal bones (5)

- These are long bones that lie in the middle part of the foot.
- They connect the tarsal bones to the toe bones.

(3) Phalanges (14)

- These form the toes.
- Each toe has 3 phalanges except the big toe which has 2.

Functions of Lower Limb :

- Supports the body weight.
- Helps in walking, running, jumping and maintaining balance.
- Protects internal organs of the pelvic cavity.
- Provides attachment to muscles for movement.

6. JOINTS (ARTICULATIONS)

The bones of the skeleton are joined together at places called joints (articulations). Joints allow movement between bones and also provide stability to the skeletal framework.

Functions of Joints :

- Joints hold the bones together.
- They allow movement of the body.
- They give support and stability to the body.
- They protect delicate structures.

Types of Joints (on the basis of movement)

Joints are classified into three types :

① Immovable (Fixed) Joints

- The bones are joined together so tightly that no movement is possible.
- Found in the skull bones.
- Example: Sutures between skull bones.

② Slightly Movable Joints

- The bones allow very limited movement.
- Present between the vertebrae.
- Example: Joints between vertebrae.

③ Freely Movable Joints

- The bones are joined in such a way that free movement is possible.
- Found in the limbs.
- Example: Shoulder joint, Knee joint, Elbow joint, etc.

Structure of a Synovial Joint

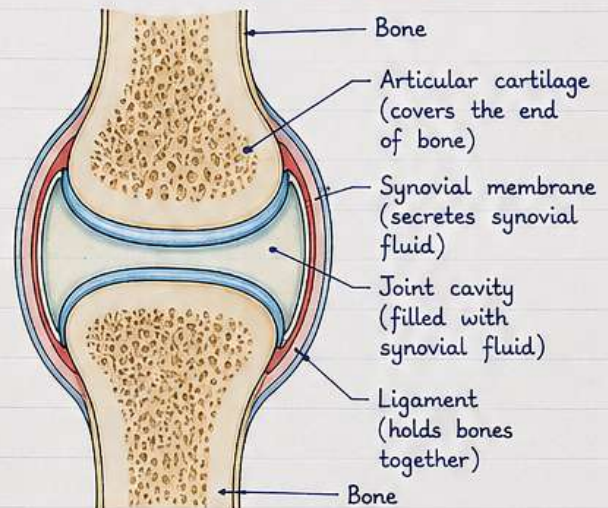


Fig. 16.19 : A synovial joint.
(Diagrammatic view)

Freely Movable (Synovial) Joints

Most of the joints in the human body are freely movable joints or synovial joints. They have the following parts (see Fig. 16.19) :

- The ends of the bones are covered with articular cartilage.
- The bones are enclosed in a fibrous capsule.
- The inner surface of the capsule is lined by a synovial membrane which secretes synovial fluid.
- The joint cavity between the bones is filled with synovial fluid which reduces friction.
- Ligaments connect bone to bone and hold the joint together.
- Tendons connect muscles to bones and help in movement.

Types of Synovial Joints (on the basis of shape & movement)

Type of Joint	Description	Movement	Example
① Ball and Socket Joint	The rounded head of one bone fits into the cup-shaped cavity of another.	Movement in all directions.	Shoulder joint, Hip joint.
② Hinge Joint	The end of one bone fits into the groove of another like a hinge.	Movement in one direction only (back and forth).	Elbow joint, Knee joint.
③ Pivot Joint	One bone rotates around another bone.	Rotational movement.	Joint between atlas and axis vertebrae in neck.
④ Gliding Joint	The flat surfaces of bones glide over each other.	Sliding or gliding movement.	Joints between the carpal bones of wrist and tarsal bones of ankle.
⑤ Saddle Joint	The saddle-shaped surface of one bone fits into another.	Movement in two directions.	Joint of the thumb.